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Physiotherapy and Osteoporosis

A guidebook for patient's education

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ABSTRACT

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The aim and objective of this thesis are to improve patients overall knowledge on osteoporosis. As this work is also to be read by healthcare and future healthcare professionals (my opponent, supervisor teacher and client), this thesis will be in two parts: the thesis "itself" (mostly for my peers and teachers) and the guidebook (for my client). Both of those parts will be focused on a physiotherapy point of view.

Even though less detailed in the guidebook, current nutrition knowledge was much developed during the thesis. Indeed, while less obviously related to physiotherapy field at first view, it is actually a major element to associate with exercises levels, in order to ensure sufficient energy levels and adaptation possibilities. Insufficient or improper energy intakes, which are common in elderly, might impact negatively bone health, especially bone mineral density, when associated with an active or inactive lifestyle. Therefore, patient education on this point was deemed as important in the thesis and guidebook.

PREFACE

In France, private physiotherapy sessions can be as short as 20 to 30 minutes, compared to an average of 30 to 45 minutes in Finland.

That time being short, physiotherapists will not have a lot of possibilities to do specific and intensive bone training with osteoporotic patients.

Some French people are lacking both knowledge and confidence to manage their pathology at home, and most are relying exclusively on the medications and diet supplements prescribed by their doctor.

This thesis and guidebook aim at providing further patient education, as well as guidance on the easiest and safest home exercises possible, so that during physiotherapy sessions harder trainings can be realized under supervision, allowing to attain more health benefits.

With special thanks to Mr. Thomas Bosek, a French physiotherapist, without whose support this thesis would not have been created.

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LIST OF SYMBOLS AND TERMS

WHO = World Health Organization

IOF = International Osteoporosis Foundation

ROM = Range Of Motion

Mesenchyme = tissue from which are originated almost all connective tissues.

BMD = Bone Mineral Density

BMI = Body Mass Index

BMC = Bone Mineral Content

PA = Physical Activity

CTXI = c-terminal telopeptide of type I collagen

DXA = dual-energy X-ray absorptiometry

FNBMD = Femoral Neck Bone Mineral Density

RM = Repetition Maximum

1 INTRODUCTION

Bone is a major element of human's body. Indeed, the adult human skeleton consists of 206 bones, each having its own name. (Tortora & Derrickson, 2017, chapter 7.1, p. 167.)

Nowadays, even though knowledge on osteoporosis among healthcare and scientific fields has drastically increased, and that many treatments have seen the day or are being under trial, osteoporosis is still a major concern affecting an important part of the population. Indeed, according to a systematic review and meta analysis published in 2021, osteoporosis would touch 21,7 % of elderlies worldwide. This study also demonstrates that the gender ratio is however not equal, with among the previously stated number women representing around 35,3 %, while men representing 12,5 %. As the population life expectancy keeps on expanding thanks to quality of life and overall progress, it is reasonable to expect an increase in those percentages. (Salari et al., 2021.)

To fight against the osteoporosis burden, one of the techniques is obviously prevention, which consists in identifying the risk factors, undergoing medical screenings and examinations, as well as being educated on the importance of diet, and age related exercising depending on several guidelines. Even though prevention is effective, not everyone, for diverse reasons including lack of education, of money, of time, etc., is able to apply it; in supplement, even in the most optimized scenario, in which each person would be able to put prevention in place, it unfortunately does not offer a 100% chance of success, and osteoporosis might still befall on individuals, regardless of the efforts furnished. (International Osteoporosis Foundation, 2025.)

This previous point brings to the topic of this thesis, which is to explore the different options to manage osteoporosis after diagnosis, including some that could be done independently in daily life by patients, and would therefore allow a little pressure release on society impact, while allowing for patients to be

better included as active participants in their health, as well as to promote their wellbeing on both psychological and physical aspects (Javier et al., 2025).

This topic is important, because the average age expectancy keeps increasing thanks to an overall better life quality (including healthcare progresses). Therefore, the worldwide amount of older people is expanding and is expected to be of 12 billion no later than in 2025. When aging, people are more vulnerable to several diseases, osteoporosis being the most common metabolic one. (Salari et al., 2021.)

Based on all the previous informations on prevalence, and expectations of increases in this last one, the current percentage of people having osteoporosis will most likely increase in the coming years (Papadopoulou et al., 2021).

2 AIMS AND OBJECTIVE

The aims of this thesis are to improve patients overall knowledge on osteoporosis.

The objective of this thesis is to create a guidebook, which consist of a summary of the theory parts detailed in the thesis, as well as on an illustrated and clear osteoporosis management advices. It will be realized on a power point support (in order to be both easy to print or to read on a phone/computer depending on the client's preferences). This guidebook will be realized and written both in English and in French.

3 OVERALL BONE HEALTH AND BONE KNOWLEDGE

3.1 Overview of bones

Together, bone and cartilage are composing what is known under the term of skeletal system. Osseous tissue has the capacity of perpetually “remodeling the bone”, meaning that aged bone tissue is broke down by the body in order to be replaced by new bone tissue. For this exact reason, bone in itself is not a simple static entity, but actually a compound dynamic living tissue constituted of many various tissues, which are: osseous (bone) tissue, cartilage, dense connective tissue, epithelium, adipose tissue and nervous tissue. In addition to this, bone is not only under one unique aspect, but can actually be classified in different types, which will ulteriorly be developed and described more precisely. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapters 6 and 6.1, p. 147.)

However, it is thanks to this, or more precisely to “long bones”, that we owe the possibility of a macroscopic observation and study of bone structure. At macroscopic level, a long bone structure is as follows: diaphysis, epiphyses, metaphyses, articular cartilage, periosteum, medullar cavity and endosteum. The diaphysis forms the bone body or shaft. It possesses an allonged cylindrical shape, and is the biggest part of the bone. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The epiphysis is used to refer either to the proximal or distal portion of a bone (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148).

The metaphysis designs a region located between the two elements previously cited, more precisely the diaphysis and epiphysis. It is on this part that can consecutively be observed (depending on the stage at which the bone is, so whether it is still in growing phase or has reached maturity) the epiphyseal plate or growth plate, ultimately replaced by the epiphyseal line (usually around the ages of 14 to 24). (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The articular cartilage refers to the thin layer of hyaline cartilage located on top of the epiphysis, at the junction with another bone. The role of the articular

cartilage is to diminish the friction and diminish shock impacts at freely movable joints. However, the fact that both perichondrium and blood vessels are lacking in that zone result in limitation in repairing if damage occurs on it. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The periosteum consists in a strong connective tissue sheath in association with blood supplies. It is located on the cartilageless parts of the bone. The periosteum may be divided into two layers: outer fibrous (dense irregular connective tissue) and osteogenic (composed of cells). The periosteum is linked to bone thickness regulation (thanks to its osteogenic layer), but is also playing an important role in protection, healing (for instance after a fracture), nourishment of bone tissue, and allows for the ligaments and tendons to attach. It is itself linked by the Sharpey's fibers (thick bundles of collagen originating from the periosteum and travelling into the bone extracellular matrix) to surrounding bones. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The medullar cavity (also known as marrow cavity) is a sunken/gaunt of cylindrical appearance found in the diaphysis. In adults, the medullar cavity contains fatty yellow bone marrow as well as many blood vessels. Its role is to keep the bone as light as possible without making it lose its power and strength. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The endosteum is a thin membrane located along the medullar cavity. It contains one layer of cells allowing bone formation and a little amount of connective tissue. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.2, p. 148.)

The structure of bone tissue can also be examined at microscopic scale, which shows that it contains an extracellular matrix, which is itself composed by various crystallized mineral salts (55%), collagen fibers (30%) and water (15%). It is at microscopic level that the process of calcification occurs. To achieve calcification, osteoblasts (bone-building cells at the initiation of calcification process) and two of the previously cited elements, mineral salts and collagen fibers, are required. The calcification process takes place in the microscopic spaces present between collagen fibers and consists in the crystallization of multiple mineral salts, namely hydroxyapatite (a combination of calcium sulfate

and calcium hydroxide), calcium carbonate and ions (magnesium, fluoride, potassium, sulfate, etc...); those crystallized mineral salts and ions fuse together; once the accumulation of crystallized mineral salts is sufficient to fill those collagen interstices, they start to gather around the collagen fibers, ultimately resulting in a combination at the origin of bone characteristics. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.3, p. 148.)

In osseous tissue, cells are present. They all play an important role in bone building and bone overall good functioning and shape. Those cells are at the number of four: osteoprogenitor, osteoblasts, osteocytes and osteoclasts. Osteoprogenitors come from mesenchyme, and are located in the inner part of the periosteum, in the endosteum, and in the bone canals presenting blood vessels. They are unspecialized bone stem cells, and at that the only bone cells able to go through cell division, resulting in osteoblasts after completion of this process. Osteoblasts are bone-building cells. As previously described and mentioned, they are the instigators of calcification. Those cells are encircled by extracellular matrix which they produce, ultimately leading to an entrapment in their secretions, resulting in the formation of osteocytes. Osteocytes refer to mature bone cells. They are the principal bone cells and ensure of the good daily maintaining of bone metabolism (example: exchange of nutrients as well as exchange of wastes with the blood). Osteoclasts design enormous cells originating from a specific type of white cells called monocytes, more specifically from the fusion of 50 of them. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.3, p. 150.)

3.2 Different types of bones

Even though it is indeed correct to say that the human skeleton is composed of bones, it is more accurate to say that it is composed of several types of bones, including long, short, flat, irregular and sesamoid (Office of the Surgeon General, 2004; Tortora & Derrickson, chapter 7.2, page 168, 2017).

Long bones are characterized by a drastically longer length than breadth, a shaft and a variable amount of epiphyses. Their diaphyses are mostly made of compact bone tissue, while their epiphyses mostly consist of spongy bone tissue. They are slightly curved, which allows them to be stronger as this

curvature promotes evenly distribution of body's weight on different bone regions, resulting in better choc absorption. Some examples of long bones include the femur, the radius, the ulna, the phalanges, etc. Those examples also highlight the fact that many different lengths of long bones exist. Short bones, contrarely to what their names might let suppose, are actually quite well proportioned with more or less equal length and width. Except for their upper layer which is made of a thin slice of compact bone tissue, they are mostly composed of spongy bone tissue. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

Short bones shape is somewhat close to the one of a cube. A significant amount of carpal bones can be cited as short bones examples. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

Flat bones are usually thin and consist of two plates of compact bone tissue surrounding a layer of spongy bone tissue. Those two plates are almost prarallel to each other. Flat bones are quite protective and allow for many muscles attachments. Cranial bones, scapulae and ribs are a few examples of flat bones. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

Irregular bones are characterized by complex morphs with various quantities of spongy and compact bone tissue. Some examples can include: the hip bone, the calcaneus, vertebrae, some facial bones, etc. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

Sesamoid bones are bones of small diameter (around a few millimeters) located in the zone of some tendons which are exposed to a lot of friction (ex: palms and soles). Due to this, they are highly variable from a person to another and are not even always completely ossified. Their role is to protect tendons by limiting wear and tear, as well as modifying the direction of pull, leading to enhanced mechanical advantage at the concerned joint. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

Those 5 bones are the main types, as almost all bones of the human skeleton can be qualified as belonging to one of these categories, but one exception, sutural bones, is also present, even thought more defining a specific location rather than a bone shape. Sutural bones are small bones which can be found in joints (sutures) between some cranial bones; how many sutural bones may be observed on the skull is highly individual and varies therefore from a person

to another. (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.2, p. 168.)

In adult body, those 6 bone categories are divided into two different parts: axial and appendicular skeleton, respectively consisting of an amount of 80 and 126 bones. The axial skeleton of the whole body consists of: the skull (8 cranium and 14 face bones), the hyoid bone, the 6 auditory ossicles, the 26 vertebral column bones, and the thorax (1 sternum and 24 ribs bones). The appendicular skeleton of the whole body consists of: pectoral/shoulder girdles (2 clavicle and 2 scapulae bones), upper limbs (2 humerus, 2 ulna, 2 radius, 16 carpals, 10 metacarpals and 28 phalanges bones), pelvic girdle (pelvic, or coxal bone, and 2 hip bones), and lower limbs (2 femurs, 2 patellae, 2 fibulae, 2 tibiae, 14 tarsals, 10 metatarsals and 28 phalanges bones). (Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 7.1, p. 168.)

3.3 Bone mass and density levels depending on age, life habits and ethnicity

The entire skeletal system is, as previously described, a living tissue that has the possibility to grow, repair and adapt.

The osseous tissue mass is influenced by several factors, such as age, lifestyle (sedentary or active), nutrition, genetics, and ethnicity (Zhu et al., 2021).

Indeed, throughout early life (more specifically during infancy, childhood and adolescence) individuals will undergo several processes such as bone growth in both length (interstitial growth) and thickness (appositional growth). Ultimately, bone peak mass will be reached, usually at around 20 to 30 years of age, and will overall stay stagnant before starting to experience a decrease, at around 50 years of age. The peak bone mass reached at around 20 years of age does not put a stop to the remodelling process. (Hereford et al., 2024; Porter & Varacallo, 2023; Tortora & Derrickson, 2017, chapter 6.5, pp. 157-158.)

As previously mentioned, some studies also highlighted the fact that genetic, as well as ethnicity, were playing a part in osteoporosis prevalence. For people with familial antecedents of osteoporosis, the chances of developing this pathology were higher compared to those who did not. (Zhu et al., 2021)

Ethny would also seem to play a part in bone mineral density according to some studies. However, the root cause remains uncertain, and it is still debated whether or not racial origins really possess an impact on bone. Indeed, according to some other studies, no major differences on Bone Mineral Density (BMD) have been observed between the different ethnical groups. Furthermore, an imbalance in participants (for instance the number of non hispanic white females was much higher in number than any other), different measurement methods and landmarks, as well as a lack of newer and bigger scale studies, also affect the reliability of current results and knowledge. Also, another limitation would reside in the need of more recent studies on the topic. (Morin et al., 2022; Noel et al., 2021.)

According to a canadian study which was interested in deepening knowledge of the hypothesis according to which origins play a role in osteoporosis occurrence (at least in Canada), the Femoral Neck Bone Mineral Density (FNBMD) would be lower in white people than in black and asian ethnies, with black persons having the higher FNBMD. In terms of fracture, white and hispanic women would both be more at risk than asian and black women. However, as acknowledged by the researchers, the percentage of white people were once again a majority (cite), and the formers were also overall older than the other participants. Furthermore, in average, the BMI of white participants was ranked as second highest after black participants, while the BMI of asian participants was ranked lowest. (Morin et al., 2022.)

Lifestyle factors are acknowledged by all sources as having an important impact on overall bone health, regardless of age and situation. However, they will be further discussed and developed in the following parts of this thesis, more specifically chapters 4.2, 5 and 6.

4 OSTEOPOROSIS

4.1 Pathophysiology, diagnosis and monitoring of osteoporosis

Many diseases can affect the skeletal system, such as for instance ÷ osteogenesis imperfecta, which is a genetic disorder due to an anomaly in the synthesis or processing of type I collagen, resulting in a lowering of bone resulting and increased chances to develop fracture. Also osteopenia, which is characterized by bone mineral density with a t-score value comprised between -1 and $-2,5$, Paget's disease, which is an osseous tissue growth disorder and osteomyelitis, a grave bone infection which can both stay acute or become chronic. osteomalacia, also called "soft bone disease", which is a condition caused by a lack of vitamin D, phosphate or calcium, resulting to an improper mineralization of bone tissue and leading to various symptoms such as lack of muscle strength, bone pain and a higher risk of fractures; bone cancers or tumors; and many others as well as osteoporosis, on which we will focus in this thesis. (Lorentzon & Cummings, 2015; Momodu et al., 2023; NIAMS, 2022; Office of the Surgeon General, 2004; Tortora & Derrickson, 2017, chapter 6.9, p. 164; Varacallo et al., 2023; Zimmerman et al., 2024.)

Osteoporosis is defined as a metabolic bone disease, and is characterized by a loss of BMD, which is itself resulting from an impaired bone microstructure. Those microstructures impairment are weakening the bone, causing it to reduce its strength and making it more prone to fractures. (Föger-Samwald et al., 2020; Porter & Varacallo, 2023; Schini et al., 2023.)

There are two types of osteoporosis: secondary and primary. Secondary osteoporosis is the developed side effect of a treatment or another pathology. Primary osteoporosis is the consequence of mechanisms related to aging. Similarly to secondary osteoporosis, it does not affect every individual. However, as previously discussed, prevalence for this pathology is high. (Föger-Samwald et al., 2020; Porter & Varacallo, 2023; Salari et al., 2021.)

Osteoporosis and osteopenia possess similarities in their pathophysiologies. Osteopenia is considered as the stage before osteoporosis and is diagnosed when t-score (an evaluation of the BMD) is comprised between -1 and $-2,5$. By opposition, the t-score needs to reach $-2,5$ or lower to be considered as osteoporosis. A normal or healthy t-score is comprised between -1 and above.

(International Osteoporosis Foundation, 2025; Porter & Varacallo, 2023; Ron-danelli et al., 2022; Varacallo et al., 2023.)

It is to a surgeon and anatomist Britishman, Sir Astley Paston Cooper, that is owed in 1822 the first observation of a correlation between altered bones and fractures occurrence. The term “osteoporosis” has been first used by Jean Lobstein, who was a French pathologist and surgeon. However, current knowledge would seem to put in evidence that the pathology to which they were referring to in their description of the condition was more what would now be known under the name of osteogenesis imperfecta type I, than real osteoporosis as understood at this day. This is not changing the fact that the term “osteoporosis” has been used since 1835 thanks to Jean Lobstein, and is still nowadays. However, a waiting time of respectively one hundred nineteen years after Sir Astley Paton Cooper discovery, and of one hundred six years after Jean Lobstein description, were needed for the basics of a hormonal treatment approach to osteoporosis to be tried and offered by Fuller Albright in 1941, after several clinical findings and observations, including menopause, loss of height in aging women, fractures, correlation with ovaries function and improvement of calcium balance with oestrogen treatment. (Lorentzon & Cummings, 2015.)

Osteoporosis is a pathology that does not only exerce a heavy weight on affected persons, but also has a big impact on society. For individuals, it is a relatively silent disease that is usually not diagnosed until a fracture has occurred, and which affects physical activity, life quality, social aspects and even life expectancy. For society, its management requires several types of means: human with caretakers and health care professionals, medical throughout medication, operations such as for instance hip replacement surgery following a hip fracture, and follow-up with the goal of checking how the fracture is healing, updating medication, physiotherapy sessions to recover strength and ROM, etc., and finally monetary as there needs to be a payment for the diverse expenses. Worldwide, the budget to cover the costs of osteoporosis is massive. In the year 2021, in the United Kingdom, Italy, France, Germany and Spain, it was revealed that the spend amount was of 29 billion euros. In United

States and in 27 European countries, it has respectively been estimated that, as osteoporosis prevalence should increase off 50% and 25% by 2025, and the osteoporosis cost would annually approach of 25.3 billion United States Dollar and 38.7 billion euros. (Papadopoulou et al., 2021.)

In France and other countries, osteoporosis is often suspected and checked for once a fracture at a typical landmark (such as the hip bone) has occurred. It is rare for the diagnosis to be made before that time. Once the suspicion has arisen, doctors will usually prescribe further examinations to the patient in order to confirm or refute the diagnosis. (Javier et al., 2025.)

Osteoporosis diagnosis and screening methods include: Dual-energy X-ray Absorptiometry (DXA), Bone Turnover Markers (BTM), and radiological assessments. However, the two last tests are more widely used in studies protocols than in a usual medical environment. (International Osteoporosis Foundation, 2025.)

While DXA is widely used, other tests can be useful indicators in some contexts, including for determining the necessity for a patient to undergo dual-energy x-ray absorptiometry or not (Bone Health & Osteoporosis Foundation, 2022).

Among those other tests are counted: Vertebral Fracture Assessment or VFA, Trabecular Bone Score or TBS, Full-length Femur Imaging or FFI, Hip Structural Analysis or HSA, etc. Those tests are not all about bone density measurements, but about finding fractures, which can also be a bone health marker. (Bone Health & Osteoporosis Foundation, 2022; International Osteoporosis Foundation, 2025.)

Regardless, DXA is one of the most precise and utilized examination means when it comes to evaluate bone health (Bone Health & Osteoporosis Foundation, 2022; International Osteoporosis Foundation, 2025).

However, more clinical assessments are also regularly checked for during patient routine visit to their healthcare provider, such as the Fracture Risk

Assessment Tool, also known as FRAX. FRAX was developed in 2008, and the goal of this computer algorithm is to determine the risk of a patient to suffer from an important fracture (at hip or other landmark) during a period of 10 years. FRAX allows to take into account several risk factors to both types of osteoporosis, for example age, gender, life habits such as smoking, BMI, medications, previous fragility fractures, family history, other pathologies, etc. FRAX is used to determine the need for treatment or further tests. (Schini et al., 2023.)

It can also be used to diagnose osteoporosis. For this, it needs to calculate an at least 20% chance of osteoporotic fracture, or an only 3% chance of hip fracture in the following 10 years. (Bone Health & Osteoporosis Foundation, 2022.)

While FRAX is not the only available algorithm, its usage is applicable to a vast majority of countries, and it is the only one that also takes into account the involved mortality risk. A more recent algorithm, FRAXplus, aiming at being more precise and reliable than the previous one, has been released in 2023. This thesis fails to gather sufficient information on it, and therefore this new screening method would benefit from further search on the topic, such as for instance the one a new thesis could collect. (Schini et al., 2023.)

4.2 Risk factors to osteoporosis

As previously explained, osteoporosis is characterized by a specific amount of BMD (Bone Mineral Density) loss.

Depending on the lifestyle (for instance sedentary), diet, medication, age, pathologies, genetic, hormones, sex and potentially ethnicity, some persons will be more likely to develop osteoporosis at a later stage (Chaudhary et al., 2024; Gkastaris et al., 2020; Metcalfe, 2008; Morin et al., 2022; Noel et al., 2021; Zhu et al., 2021).

Some risk factors to reduced BMD include: being a female, having Caucasian or Asian ethnicity, aging, low BMI (Body Mass Index), menopause, slender habitus, sedentary behavior, smoking, alcohol consumption, genetic or in other

words family history, unaproprate or insufficient diet, being under certain medications such as corticosteroids, heparin, cyclosporin, cytotoxic drugs (and many others), endocrine diseases such as Cushing's syndrome, hyperparathyroidism, hypogonadism, acromegaly, type I diabetes mellitus (and many others), joint disease such as rheumatoid arthritis, and other sickness issues, such as chronic renal failure, chronic liver disease, mastocytosis, anorexia nervosa, inflammatory bowel disease, coeliac disease, etc. Furthermore, obesity is also a risk factor, as revealed in a recent study demonstrating that osteopenia and osteoporosis prevalence would be correlated with abdominal adiposity. Childhood obesity would as well be linked with decreased bone strength. (Chaudhary et al., 2024; Gkastaris et al., 2020; Metcalfe, 2008).

Some of those risk factors and knowledge on ways to prevent them are now controversial and highly debated.

One example is obesity. While it was long believed that the fat would act as a cushion effect on bones, therefore protecting efficiently against fractures, more recent studies tend to prove that it is not the case, while some others encline to think that it is at the very least not as easy as it seems. It would for instance seem that, if obese women do seem to be less likely to suffer from fractures in the wrist, hip, and pelvic areas, they would however be more prone to them in different locations, such as the ankle, the leg, the humerus, and vertebral column. Additionally, obese persons would be at higher risk of disability and falls (including repetitive ones) in comparison to leaner persons, due to several factors such as dynapenic obesity (loss of muscle strength dur to weight issue), sarcopenic obesity (loss of muscle mass due to weight issue), etc. Obese persons would even demonstrate a different balance loss and falling pattern than from average weight individuals. Furthermore, obese persons would be more likely to fall in a backward or sideways fashion, while others would be more likely to fall in a forward fashion. (Gkastaris et al., 2020.)

4.3 Importance of overall osteoporosis knowledge and management for both healthcare professionals and patients

A study specifically centered on the french osteoporosis management, gathered patients' opinions on several aspects of it, such as their need for assistance, the quality of education and information concerning their pathologies, their trust in available treatments, the impact of the disease on their daily life, etc. The conclusions of it were that, on many, if not all of those previously mentioned points, most of the study participants noticed either a decrease in comfort and independance during their daily life, or expressed a need for better medical guidance and regret that it was not so. Some members of the study also voiced concerns regarding their ability to trust healthcare professionals, as well as in the reliability of pharmacological industries. While the importance and efficiency of drug therapy in osteoporosis treatment is undebated, it is not part of a physiotherapist role, role on which is the main focus of this thesis; therefore, the patients points of view on the interactions with healthcare professionals, as well as their physical impacts, will be more developped. A big majority of the study participants felt as the amount of informations they received on osteoporosis were insufficient, and as if they were not able to discuss of the potential management techniques. Some participants also seemed to consider worsening of the condition unavoidable, and anxious about its evolution and changes following from it (such as future purchase of a one floor house, etc.). Except for one female participant who did not notice differences in her physical capacities, all experienced anxiety on exercising, and became more relying (for instance to carry loads, gardening and doing chores) on their relatives, mostly their husband, or sometimes on paid staff. Building better communication and trust between healthcare professionals and patients should be aimed at, in order to achieve better results on osteoporosis treatment, allowing at the same time to relieve both sides from the burden of this disease. Furthermore, support and ideas on home based coping managements should be given, and those coping techniques should be encouraged to enhance patients comfort in daily life. This thesis derived product, the guidebook, will therefore aim at responding to those needs. (Javier et al., 2025.)

5 FACTORS INFLUENCING OSTEOPOROSIS AND OVERALL BONE HEALTH

5.1 Nutrition

Several factors are able to influence on overall bone health and can participate in prevention of BMD loss. However, as previously explained, prevention techniques are not always sufficient, due for instance to genetical predisposition, or even able to be put in place because lack of money, time, or education on its importance and techniques, etc. (International Osteoporosis Foundation, 2025.)

Even so, it is still possible to slow down the BMD loss process, improve fracture recovery as well as prevent new ones, lower the incidence of falling and participate in musculoskeletal health, with the help of several means, for instance medication, physical activity and nutrition (International Osteoporosis Foundation, 2025).

Nutrition is defined by WHO as the amount of food a person is taking, in comparison to that same person's dietary needs (WHO, 2025).

While appropriate amounts of food are required in order to live, it is important healthwise to ensure that good nutrition is received, as it plays a continuous key role in human proper development from life to death. Insufficient nutritional intake may lead to lowered immunity, higher chances to develop diseases, repercussions on mental and physical development, and lessened productivity. On the other hand excessive nutritional intake increases overweight and obesity chances, therefore also leading to higher chances of suffering from strokes, heart and other cardiovascular diseases, diabetes (type II), cancer (breast, endometrial and colon), physical disability, and musculoskeletal issues (for instance osteoarthritis). (Gkastaris et al., 2020; WHO, 2025.)

Inappropriate food intakes are linked, as previously described, with bone health, and more precisely BMD (Gkastaris et al., 2020; International Osteoporosis Foundation, 2025; Kanis et al., 2018; Metcalfe, 2008).

Several studies were made in order to assess the connection between nutrition and bone condition, as well as whether there was actually an existing one. More and more studies demonstrate that appropriate diet has effects on the remodelling of bone, on the mineral characteristics of bone, as well as on BMD. While many of those studies are focusing on adults population categories, one study underlined the potential impact of nutrition on present and future bone health for children and adolescents, as well as the lack of individual research for those last ones, whose guidelines are almost exclusively based on adults. (Kędzia et al., 2023; Papadopoulou et al., 2021; Proia et al., 2021.)

A study on the impact of physical activity and nutrition on sarcopenia and osteoporosis gathered informations on dietary recommendation for persons suffering from BMD loss. Contrarely to a very wide assumption, administration of vitamin D on its own to a population presenting no deficits to it, as well as the ones of dairy and milk products, are not associated with benefits on osteoporosis prevention nor chances and recurrence of fractures. (Kanis et al., 2018; Papadopoulou et al., 2021.)

When aging, people tend to have a decrease in their nutritional intake, leading more easily to BMD loss and other health issues. Regardless, a qualitative food intake, even reduced, has shown positive impact on functional capacity and muscle strength in elderlies population. This was particularly true for the Mediterranean diet, which contents are high in healthy fats, fish, fruits and vegetables. (Papadopoulou et al., 2021.)

An intake of many various macronutrients (such as vitamin K (MK-7), vitamin D, copper, calcium, magnesium, zinc and manganese), but also of fluorid and strontium, was found to be important on overall bone health. Fluorid and strontium would have proven themselves essential to stimulation and inhibition of osseus tissue cells, respectively osteoblasts and osteoclasts. (Chaudhary et al., 2024; Papadopoulou et al., 2021.)

As previously mentioned, vitamin D as a monotherapy did not demonstrate itself as an efficient component, but a study on a group suffering from sarcopenia underlined that muscle gain was observed when vitamin D and protein intakes were associated. Furthermore, protein intake would seem to play a

protective role in what comes to lumbar spine bone density. (Kanis et al., 2018; Papadopoulou et al., 2021.)

However, it is to remember that protein intake, in order to have maximized efficiency, needs to be adapted to the level of physical activity, as the amount recommended in diet will vary whether the person is at bed rest or physically active. This is reminded in a Finnish study which demonstrated a correlation between excessive protein consumption and decrease in BMD and Bone Mineral Content (BMC). Protein should also need to come from varied sources (animal and plant based). Protein has been shown to be one of the most important macronutrients for bone health, and a lack of proper protein intake in diet results in a faster bone mass loss, when compared for instance to a dietary regimen with insufficient amount of vitamin D. Indeed, protein would represent 50% of bone volume, as well as 1/3 of bone mass. Nevertheless, the role of protein is debated. Protein can act as a builder of bone mass, play a part as hormone and collagen syntheses, have an anabolic effect through the expansion of insulin-like growth factor 1 (IGF-1) and bone growth. On the other hand, protein can also be linked with bone health impairment through the worsening of calcium excretion in persons suffering of renal dysfunction. Furthermore, an excessive protein consumption leads to increased c-terminal telopeptide of type I collagen, marker which worsens the loss of bone mass, and that regardless of other nutritional intakes, such as potassium and calcium. Still, when part of a properly balanced diet allied with adapted dosage depending on level of physical activity, it would seem beneficial. Even though nutrition does play a key role in bone and even overall health, the best results on BMD are observed only when appropriate nutrition (based on age, gender, level of Physical Activity (PA) and presence of co-existent health conditions) is combined with physical activity, and in some cases medication. It is therefore safe to assume that the action of nutrition itself is limited, and will not be sufficient for overall bone health, including specific conditions such as this thesis topic, osteoporosis. (Kędzia et al., 2023.)

Furthermore, improperly dosed nutrition in comparison to a person's level of physical activity, regardless of whether that person is active or not, showed negative impacts on bone density; also, persons with insufficient energy levels undergoing physical trainings of high intensity and volume, demonstrated

negative impact on bone formation, as well as increased chances of menstrual dysfunction (Papadopoulou et al., 2021).

5.2 Medication

Since the discovery of osteoporosis, treatment protocols, based on clinical observations, started to be experimented. Nowadays efficient drugs and other forms of therapy (such as hormonal) have been developed. (Kanis et al., 2018; Lorentzon & Cummings, 2015.)

In Europe, several medications aiming at reducing the risk of fractures, are used and prescribed to patients. Among them, the most common are: raloxifene, bisphosphonates alendronate, clodronate, ibandronate, risedronate and zoledronic acid, which are agents obtained from parathyroid hormone and denosumab. While all those medications have been found as efficient to lower the incidence of fractures at the vertebral spine, only few of them have been recognized as effective for other landmarks (for example bazidoxifene, alendronate, risedronate, ibandronate, teriparatide, denosumab, clodronate), such as for instance the hip bone (for instance alendronate, risedronate, zoledronic acid, denosumab). However, not all European countries allow all of those drugs on their individual territories. Indeed, an example could be bazidoxifene, which can only be prescribed in Spain and Germany. Therefore, not all the previously cited medications can be considered as relevant in this thesis, which is centered on France. Furthermore, while it was previously mentioned that those drugs can not prevent fractures from each location, it is also to be remembered that, due to potential side effects and counter indications, the benefit risk of treatment needs to be evaluated by a doctor and discussed on with the patient. Indeed, some medications can increase the risk of deep venous thromboembolism (even though extremely rarely in the case of raloxifene), strokes, oesophagitis, as well as possibly arrhythmia. Lesser side effects are as varied as hot flush, cramp in inferior limbs, gastrointestinal disturbances, nausea, fever and muscle/bone pain (temporary after an intravenous administration of amino-bisphosphonates), headache, dizziness, etc. (Kanis et al., 2018.)

On the other hand, some side effects such as osteonecrosis of the jaw, while a big concern of patients undergoing biphosphonates treatment, are actually extremely rare and would yearly only concern 1 out of 100 000 patient. It is a bit more unclear whether prolonged usage would increase those risks, but the class of biphosphonates would also seem to play a part in this. (Javier et al., 2025; Kanis et al., 2018.)

To counter some of those side effects, other drugs are used in association, but those were found to reduce at the same time the bones benefits. Also, some medications efficiency was impaired by some types of foods, making them restrictive during daily life and requiring a diet monitoring. Furthermore, the beneficial effects of some medicines, like denosumab, stop shortly after discontinuation of the treatment, and losses in BMD can be seen in no less than 12 to 18 months time. (Kanis et al., 2018.)

Still, medication based treatment, when associated with proper exercising and nutrition, did impact positively the well being and bone health of patients (International Osteoporosis Foundation, 2025; Kanis et al., 2018).

5.3 Physical activity

It is of common knowledge among all healthcare professionals, as well as among most people, that regular physical activity and appropriate exercises (dosed, etc.) present numerous health benefits on both body and mind (WHO, 2024).

The WHO established physical activity guidelines for different population categories, divided up by age. Osteoporosis being mainly affecting people starting from around 50 years old (and over) for women and from around 65 years old (and over) for men, the most relevant activity guidelines for this thesis are the following two: 18 to 64 years old, 65 years old and older. (NIAMS, 2023; WHO, 2020; WHO, 2024.)

For the first previously cited age group, it is recommended to adults to do, per week, no less than between 2h30 to 5h of moderate intensity aerobic practice, and between 1h15 to 2h30 of vigorous intensity aerobic practice. More benefits

are observed if persons also get involved in muscle strengthening practice for at least 2 days per week. In some cases, increasing those amounts of physical activity can be beneficial, but depending on more individual parameters. For the second previously cited age group, the weekly recommendations are no different than for the first one, at the difference that multicomponent trainings involving strength and functional equilibrium should be performed at least for no less than 3 days a week. Increasing those amounts of training may one more time show benefits, but remains dependable on situation and individual cases. However, it has been underlined by recent studies that 31% of world-wide population is physically inactive, with existing disparities between males and females of 5% since 2000, the formers being less likely to engage, at least sufficiently, in physical activity. Furthermore, after the age of 60 years old, the population, regardless of gender, would present higher levels of physical inactivity, even though its importance and benefits are by no means reduced on aging individuals. Therefore, patient information and education are essential in order to reverse this tendency. (WHO, 2024.)

Studies revealed correlations between bone health, osteoporosis and physical activity. Those studies are often closely intertwined with nutrition. (Papadopoulou et al., 2021.)

Recommendations guidelines for osteoporosis usually consist in moderate-intensity training, that is 70-80% 1 RM, 8-15 reps, and should therefore be considered when prescribing activity to patients, in plus of the WHO age based recommendations. It is furthermore for physiotherapists that physical activity is the most important, both as a topic on which educate the patients, but also as a treatment, knowing physical activity is the main osteoporosis treatment method used in physiotherapy field. (Watson et al., 2019; WHO, 2024.)

However, rather than physical activity, which is a large topic, targeted exercises are deemed more beneficial for bone health, and conditions such as osteoporosis. Therefore, it is essential for physiotherapists to be as

knowledgeable on exercising techniques and their various recommendations, than on physical activity and its dosage. (Hong & Kim, 2018; Manaye et al., 2023; Ng et al., 2023; Papadopoulou et al., 2021; Watson et al., 2019.)

6 DIFFERENT TYPES OF EXERCISES, CONCEPTS OF ADAPTATION-PROTECTION AND MECHANICAL STRESS QUANTIFICATION

6.1 Specific exercises

As previously mentioned, physical activity is tightly correlated with bone health regardless of the age group (children and adolescents, adults and elderlies) and associated pathologies, such as this thesis topic, osteoporosis, but also sarcopenia and osteopenia (Papadopoulou et al., 2021).

However, for further and more targeted benefits on osteoporosis, specific exercises programs for bone proved to be beneficial, more specifically at decreasing the bone loss in terms of mass and strength, reducing as well the risk of fractures (Hong & Kim, 2018; Manaye et al., 2023; Ng et al., 2023; Papadopoulou et al., 2021; Watson et al., 2019).

Among the recommended exercises, balance (in association with functional training), resistance and weight-bearing are deemed beneficial for both prevention of falls and management of osteoporosis. Concerning functional and balance training, it has been shown to decrease the likelihood of falls among fracture prone individuals by 24%. (Bae et al., 2023; Giangregorio & Ponzano, 2022; Papadopoulou et al., 2021; Watson et al., 2019.)

Still, due to an heterogeneity in some of the researches results, further studies would be needed (Bae et al., 2023; Papadopoulou et al., 2021; Watson et al., 2019).

This heterogeneity might be explained by the lack of sufficient loading on the bone during weight-bearing. Indeed, low impact activity, such as walking, swimming or cycling which, while possessing numerous other benefits for cardio, mobility and muscle mass, would not be sufficient to make the bone feel a necessity to adapt to the change in activity. Higher impact activities, on the other hand, would be beneficial only on certain cases, in a progressive manner, and under supervision due to the enhanced chances of improper realisation and injuries. (Manaye et al., 2023; Papadopoulou et al., 2021; Schinzel et al., 2023; Watson et al., 2019.)

Among the higher impact activities program, one of them, named HiRIT, which stands for High-intensity, progressive Resistance and Impact weight-bearing Training, was tested in a LIFTMOR studies. In this study, participants suffering of either osteopenia or osteoporosis, were coached two times a week during 8 months throughout a HiRIT protocol, more precisely 5 sets of 5 repetitions, >85% 1 repetition maximum. In the same study, the control group was undergoing a low-intensity home program, consisting of 10 to 15 repetitions at <60% 1 RM. While during the last BMD measure by DXA, the HiRIT group was only having slightly better results than the control group, it however demonstrated a much more positive impact on posture, resulting in an increase in length and reduced kyphosis. However, the study was having a relatively small number of participants, and while none of them was ever endangered by training, not only positive effects on bone health were noted, as 18,6% of participants experienced a decrease in their lumbar spine BMD, and 28,8% in their femoral neck BMD. Still, this number was much lower than in the control group, where 72,1% of participants experienced lumbar spine BMD loss, and 62,8% in their femoral neck BMD. (Watson et al., 2019.)

While debated, aquatic high intensity exercises programs were correlated by a study from 2013 with an increased amount of marker P1NP and lessened amount of markers for bone resorption, while other researchers found no or very little beneficial impacts on bone health (Moreira et al., 2013; Papadopoulou et al., 2021).

More recent studies also interested themselves to the effects of water exercises for geriatric or postmenopausal groups, as well as for elderlies suffering of osteoporosis and treated with medication. Those three studies, while underlying different conclusions on whether bone formation would be positively impacted by aquatic exercises or not, all reached the conclusion that it was efficient in lowering and slowing down the bone resorption, therefore providing bone health benefits. Even though more research is required, aquatic based training was still deemed safe, more enjoyable and more reassuring for participants, especially those with fear of falling and impairments (such as more balance issues for example). Those conclusions were reached including in conditions where participants had to jump in water. Depending on the gravity of osteoporosis and of course on the participant specific needs and conditions, aquatic exercises could be used either as a first step towards higher intensity ground based progressive activity, and or as a complement to it. While agreeing on the previous points, the three researches highlighted different benefits: the study by Aboarrage et al., (2018) focused more on positive effects on overall fitness and BMD, while the study by Lv et al., (2021) focused more on the augmentation of positive bone markers, while finally the study by Schinzel et al., (2023) focused more on the significant BMD gains on specific landmarks, namely the femoral neck and lumbar spine. However, while there is undeniably a mechanically induced load on the muscles during water exercises, maintaining a more on ground training approach where loading would also be linked to gravity, remains an important component of bone homeostasis proper functioning, and should therefore not be forgotten or put aside. Overall limitations on the specific actions and conclusions regarding the benefits of aquatic exercises for osteoporosis, discussed in this section, would result from the need of more studies, the continuous researches on exercises topic, the necessity where some researchers were of translating certain studies used among their cited sources (ex: from Japanese to English), heterogeneity of the studies protocol, lack of specificity regarding dosage and intensity of exercises, lack of clarity on blinding of participants and therapists, methodology weaknesses, and some different outcomes on whether certain types of exercising had adverse or positive effects. While this does not invalidate the conclusions of those studies, or their strengths on other parameters, nor the interest they represent for the overall research on the water exercises topic, it is not either to forget

that higher scale studies investigating the question under same angle, would be beneficial in determining more accurately the extent of benefits on bones health, as well as the optimum conditions (height of the water, water temperature, training and session durations, etc.) of the program. (Schinzel et al., 2023; Lv et al., 2021; Aboarrage et al., 2018; Kemmler et al., 2023.)

6.2 Adaptation-protection concept

While targeted exercises possess a highly beneficial impact on bones health, as well as on enhancing bone capacities such as repairing, strengthening, etc., it is primordial that proper preparations and dosage are instructed to the patients and put into place by both the former and the physiotherapist. Indeed, while it is certain that physical activity as well as physical exercises are providing many advantages, they are also able to have inefficient or even reversed effects, which can be especially harmful on people whose bone health is already decreased and weakened by osteoporosis, with a suspicion of making them more prone to exercise induced osteoporotic fractures. (Watson et al., 2019.)

However, while most programs for osteoporosis are, as previously described, not enough loading to have a true beneficial impact on bones and fracture preventions, more and more evidence seems to prone a progressive but supervised and more intense program, in order to reach the objectives, or in other words the concept of adaptation (La Clinique Du Coureur, 2024; Watson et al., 2019).

Adaptation is closely linked to protection (La Clinique Du Coureur, 2024).

The protection concept is to minimize movements, loads and impacts by several means, such as rest, orthotics and exclusive use of manual therapies. While it is useful during the acute phase of an injury (for instance a fracture) to manage symptoms such as pain and inflammation, it usually demonstrates negative results at long-term, as it can fragilize soft tissues, diminish tolerance to activities or exercises. It could even result for elderlies in a BMD loss

equivalent to one year of bedrest for a healthy individual, and that in no longer than one week. (Kanis et al., 2019; La Clinique Du Coureur, 2024.)

The adaptation concept is to progressively use movement as a reinforcement and treatment method, while introducing cardiovascular activities at the same time. While the progression needs to be closely monitored and controlled to avoid short-term risks, such as injuries and/or soft tissues irritation, it has been linked with increase in BMD, as well as overall fitness level and resistance of soft tissues. (Hong & Kim, 2018; La Clinique Du Coureur, 2024; Manaye et al., 2023; Ng et al., 2023; Papadopoulou et al., 2021; Watson et al., 2019.)

The safety of adaptation concept is itself based on a proper mechanical stress quantification concept, which will be explained in the next part.

6.3 Mechanical stress quantification concept

The mechanical stress quantification refers to a study or an analysis which is made in order to ensure that the stress (impacts, load, etc.) applied to the body and its composing structures is sufficient to allow those to adapt themselves through reinforcement before increasing the requirements (for instance intensity), while not being too intense or demanding in order not to result in an over-use injury. In summary, this concept main point is to preserve from trauma while progressively loading the body through exercises. In order to do so, quantifying the stress amount is primordial, for as long as it is not higher than the adaptability capacities, injuries should be prevented. However, several factors, not all physical (or at least not purely), are to take into account. Indeed, the results of mechanical stress quantification analysis and the recommendations following from it are far from being static, and are actually impacted by a person's overall state. For example, if a person is tired, was recently sick, or is anxious, all those will affect on how much "impact", "load" they will be able to tolerate during exercises. Furthermore, injuries (during and post care) or

pathologies are also to take into account and usually require opinion and advice from a healthcare professional. (The Running Clinic, 2025.)

Throughout this thesis and guidebook, the mechanical stress quantification concept will be adapted and targeted to people suffering of osteoporosis, but it is to remember that if other diseases or traumas are associated, then it is preferable to get even more individual guidance from your usual doctor and/or physiotherapist. The same recommendation applies to persons who would express doubts on the benefits or risks that exercises recommended in the guidebook could pose to them, as it based on general guidelines and can therefore not be too individualised. However, as an important component of the guidebook is to remain safe, so as to meet the objective of allowing patients to use it for self management, the exercises will be divided by level of difficulty and intensity. Programs such as HiRIT, while deemed beneficial, will be left to the judgment of the healthcare professional following the patient and excluded from this work, as the recommendation regarding this type of training is to be supervised and closely monitored. (Hong & Kim, 2018; Manaye et al., 2023; Ng et al., 2023; Papadopoulou et al., 2021; The Running Clinic, 2025; Watson et al., 2019.)

The relationship between all osteoporosis main key components discussed previously in this thesis will be reused in chapter 8 as well as in the guidebook. The guidebook itself will be more focused on in the next part.

7 GUIDEBOOK FOR OSTEOPOROSIS SELF-MANAGEMENT

7.1 Summary of theory parts on osteoporosis

The aim of this guidebook being to be at the same time instructive, clear and concise, choices were needed to be made during its redaction.

Indeed, while some patients may definitely be able to understand the content of this thesis as itself, and be curious or disposing of enough free time to read it in its entirety, it is most definitely not the case for everyone (Javier et al., 2025; Watson et al., 2019).

Therefore, a short definition of osteoporosis, the descriptions of few osteoporosis signs that can be self-checked (for those who would wish to educate their close ones on screening, or are simply not sure of their own condition), an enumeration of management techniques, an explanation on the benefits of physical activity, and finally a few safe home exercises, will be included in the guidebook.

To create the guidebook, knowledge was gathered in this thesis through thought searches on various sources and databases, such as: Finna, Pedro, PubMed, google scholar, NCBI, books, as well as (but more occasionally) other websites such as the World Health Organization (WHO) and International Osteoporosis Foundation (IOF). All sources are traceable through rigorous in-text citations and references in both the guidebook and thesis. However, the practice-based part of this work, the home-based exercise program of the guidebook, was based on training protocols from scientific studies as well as previous work placement experience with patients who had low levels of physical activity or had osteoporosis; those training protocols were modified and are therefore not similar, but they were still the foundation of the home-based exercise program, in order to ensure safety and evidence according to current official recommendations.

7.2 Time of action of the different exercises types and overall evidence based recommendations

Several studies on elderly populations, as well as on patients suffering of osteoporosis, were able to notice exercise-induced positive changes in bone density at around 6 or 8 months. Indeed, while it takes 4 months for a bone remodeling cycle to occur, the changes are not immediately visible on imagery, and it is better to wait at least 2 months, sometimes even a little longer. While changes in bone health occur slowly but surely, and need to be monitored with specific tests to be seen, some other interesting benefits of exercises, such as

for instance balance, occur at a faster rate and are noticeable by patients, even after only a month or so. Being able to realize the changes by themselves might be encouraging to them, but could also promote their participation and confidence, for example by reducing their fear of falling. Regular exercising would also give more control to patients over their pathology, while at the same time allowing healthcare professionals, especially physiotherapists, to do more intense, effective and challenging sessions with their patients, which are currently pointed out by literature as being more effective. (Watson et al., 2019.)

7.3 Complete patient guidance through the guidebook

Throughout the guidebook, choice was made to guide the patients suffering of osteoporosis with exercises and nutrition, while other treatments methods, such as medication and dietary supplements, were simply mentioned but not discussed.

This exclusion was decided on the basis that drugs can only be prescribed by a registered doctor, and that literature strongly recommends a physician opinion on the needs and requirements of dietary supplements in order to ensure that no negative side effects will occur on health (International Osteoporosis Foundation, 2025; Javier et al., 2025).

Physical activity and exercises were more focused on, as they have been proven to achieve good results, and are obviously the most physiotherapy centered approach to osteoporosis management (Papadopoulou et al., 2021; Watson et al., 2019).

Nutrition was as well, even though briefly, explained to patients along with some diet recommendations. While less clearly related to physiotherapy, its close intertwiness with exercises, and the undeniable impacts regarding the association of both on bones health, made it an important part of patient's necessary education. (Papadopoulou et al., 2021.)

8 DISCUSSION

During this thesis and guidebook processes, no ethical issues related to anonymity and confidentiality were present, as the gathered data, guidelines and informations came from previous published studies on the osteoporosis topic, and not from participants with whom direct contact would have been established. Concerning the client of the guidebook, his consent, implications and roles were explained and defined before any official agreement took place.

However, ethical issues might reside in the fact that a majority of participants, at least in some of the mentioned studies, were women, and therefore men and the effects of current treatments more specific to them might be underrepresented; still, it is to note that no gender based differences in treatment seemed necessary in the guidelines of physical activity. Furthermore, in regard to the guidebook, a difficulty to balance efficacy and safety in the exercises while adapting them so as to be feasible for almost anyone, might not be reached in all cases. Also, while rare, adverse effects of physical activity for more intense workouts, or insufficient effects for the most gentle ones, have been observed on some patients, and while this point has been clearly explained in the guidebook along with the necessity to consult a physiotherapist for more challenging exercises, they might as well represent an ethical issue, even though they are in perfect concordance with the recommendations, which are to keep intensive training for supervised session, and gentler exercises for safe home usage. Finally, there is always a possibility for informations to be misunderstood or confusing, which could be problematic for patients exercising alone at home; it is in order to avoid this situation that recommendations for patients to seek advice from their regular physiotherapist or doctor if deemed safer or reassuring is expressed in the guidebook.

Indeed, the management of osteoporosis is complex and involves both patients and healthcare professionals from different fields (nurses, doctors, physiotherapists, rheumatologists, etc...). It also involves several treatment options: medication, nutrition and physical exercise. (Javier et al., 2025; Kanis et al., 2018; Papadopoulou et al., 2021; Salari et al., 2021.)

Osteoporosis, while being able to improve with proper management, remains an incurable disease that can not always be fully avoided by prevention (Föger-Samwald et al., 2020; Papadopoulou et al., 2021; Porter & Varacallo, 2023; Schini et al., 2023; Watson et al., 2019).

Therefore, its weight on society and patients, the formers being often overpowered by lack of education and /or flow of information, is massive (Javier et al., 2025; Papadopoulou et al., 2021; Salari et al., 2021).

To avoid kinesiophobia, which too often results in an even faster and important BMD and muscle loss, but also in higher chances of falls and physical capacity diminution, proper exercises guidelines and home programs would be useful to patients (Javier et al., 2025; Watson et al., 2019).

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